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BSIT-4E

ABC

STARTUP

SOLUTIONS

Enterprise IT Infrastructure Plan

Executive

Summary

This document presents the proposed IT infrastructure for ABC Startup Solutions, a fictional software development company with 20 employees working on a single office floor. The company is starting without computers, servers, network equipment, internet infrastructure, or formal security policies. The plan establishes a structure that can support daily office operations, software development, internal services, data protection, and future expansion. The proposed environment uses Windows 11 Pro for employee workstations, Ubuntu Server LTS for central server services, a managed 48-port switch, separate routing and firewall functions, three wireless access points, NAS-based local backup, and offline backup drives. Department-level VLANs are recommended to separate business traffic and make security rules easier to manage.

PART 1 - COMPANY PROFILE

Profile Item	Details
Company Name	ABC Startup Solutions
Nature of Business	Software development, web and business application services, system integration, and technical support.
Company Vision	To become a trusted technology partner for growing organizations by delivering practical, secure, and dependable.
Company Mission	to develop useful software and technology services while maintaining reliable operations, protecting client information.
Office Location	Unit 4B, Meridian Business Center, National Highway, Los Baños, Laguna, Philippines
Organization Type	Startup software development company with a single office floor and a centralized IT function.

Organizational Structure

The company uses a simple functional structure. Management oversees the overall business direction, while the four departments handle their respective responsibilities. Information Technology provides technical support and infrastructure administration to the whole organization

Department	Employees	Primary Function
Information Technology	5	Application development, infrastructure support, user support, and technical operations.

Human Resources	4	Recruitment, employee records, attendance, and personnel administration.
Finance	5	Accounting, budgeting, payroll coordination, and financial records.
Sales	6	Client communication, proposals, sales activities, and customer follow-up.
TOTAL	20	

Planning Assumptions

The design assumes a single-floor office, one main communications/networking area, normal business-hour operations, wired connections for fixed workstations, business-grade Wi-Fi, and a requirement to keep critical company files backed up locally and offline. The plan also leaves spare capacity so the company can add employees without replacing the core network immediately

PART 2 - ENTERPRISE HARDWARE INVENTORY

Asset ID	Hardware	Quantity	Department / Area	Purpose / Justification
HW-001	Business Desktop Computer	20	All departments	One primary workstation per employee. Standardized desktops simplify imaging, maintenance, and replacement.
HW-002	Business Laptop	6	IT / Sales / Management	Provides mobility for IT administration, meetings, presentations, remote work, and contingency replacement.
HW-003	Rack Server	1	IT / Server Room	Runs central services such as file services, internal applications, monitoring, and selected virtual machines.

HW-004	48-Port Managed Switch	1	Server/Network Room	Provides enough wired ports for 20 users, server, printers, access points, uplinks, and future growth.
HW-005	Business Router	1	Network Room	Provides WAN routing, gateway services, and controlled traffic between the office network and ISP.
HW-006	Network Printer	2	HR / Finance / Common Area	Two units reduce walking distance and provide a backup printing option if one unit is unavailable.
HW-007	UPS	4	Server/Network/Finance	Protects server, network equipment, NAS, and one critical office area from short power interruptions and voltage problems.
HW-008	Wireless Access Point	3	Office Floor	Provides business Wi-Fi coverage while allowing separate staff and guest wireless networks.
HW-009	NAS Storage	1	Server Room	Provides centralized local backup storage and a secondary location for important files.
HW-010	External Backup Drive	2	IT / Secure Storage	Used for offline backup rotation. Keeping one drive disconnected reduces exposure to ransomware

				and accidental deletion.
HW-011	24-inch Business Monitor	26	All departments	Provides one display per desktop plus additional displays for IT and selected laptop users who need dual-screen work.

Hardware Selection Rationale

The inventory is intentionally standardized. Twenty desktops cover the 20 employees, while six laptops are reserved for roles that benefit from mobility and for temporary replacement use. A 48-port managed switch is preferred over a small unmanaged switch because it provides room for VLANs, monitoring, access points, printers, and additional users. The backup design combines NAS storage with two external drives so that not every backup remains connected to the production network

PART 3 - ENTERPRISE SOFTWARE INVENTORY

Software	Version / Release	License	Purpose / Reason
Windows 11 Pro	Current supported business release	Commercial / per-device license	Primary desktop operating system. Provides business management and security features such as BitLocker, TPM support, Windows Firewall, and account-management options.
Ubuntu Server	26.04.1 LTS	Open source; optional Ubuntu Pro	Server operating system for internal services. LTS support makes it suitable for a stable server environment.
Microsoft Office / Microsoft 365	Current supported release	Subscription commercial /	Provides word processing, spreadsheets, presentations, email, and other business productivity tools.
Visual Studio Code	Current stable release	Microsoft product license	Lightweight editor for application development, scripting,

			configuration files, and technical documentation.
Git	Current stable release	GPL-2.0	Version control system used to track changes in source code and configuration files.
GitHub Desktop	Current stable release	Open source application	Provides a graphical interface for Git repositories, commits, branches, and synchronization.
VirtualBox	Current supported release	GPLv3 / extension licensing varies	Allows IT staff and developers to create virtual machines for testing, training, and isolated environments.
Google Chrome	Current stable release	Freeware / proprietary	Web browser used for cloud applications, documentation, testing, and business web systems.
Microsoft Defender	Built into Windows 11 Pro	Included with Windows	Provides endpoint malware and threat protection for Windows computers.
AnyDesk	Current supported business release	Commercial / plan-based	Allows authorized IT personnel to provide remote technical support to users.
7-Zip	Current stable release	LGPL / open source	Used for file compression and extraction of software packages, backups, and documentation files.

Software Management Approach

LGPL / open source Remote support tool for authorized IT assistance. Access should require strong authentication and be limited to approved administrators. File compression and extraction for software packages, backups, and documentation files. All software should be installed from official sources, documented in the software inventory, and kept on a regular patch cycle. Commercial licenses should be assigned to the correct users or devices. Administrative tools such as AnyDesk should be restricted to IT personnel and used only for approved support activities. Windows 11 Pro includes several built-in business security capabilities, while Ubuntu Server LTS provides a stable base for server workloads. Microsoft and Ubuntu documentation should be checked before major version upgrades. Official documentation

confirms that Windows 11 Pro includes business security capabilities such as TPM, BitLocker, Windows Firewall, and account-management features. Ubuntu states that LTS releases receive standard security updates for five years, with longer coverage available through Ubuntu Pro. Visual Studio Code's product license permits use for development and testing, while GitHub provides GitHub Desktop as an official desktop Git client.

PART 4 - ENTERPRISE NETWORK INVENTORY

Asset ID	Network Equipment	Quantity	Recommended Specification	Purpose
NET-001	ISP Modem / ONT	1	ISP-provided fiber ONT/modem	Terminates the internet service and provides the WAN connection to the company's router.
NET-002	Business Router	1	Gigabit/2.5GbE, VLAN-aware, VPN-capable	Handles WAN routing, NAT, DHCP where appropriate, and network gateway functions.
NET-003	Firewall	1	UTM/NGFW with VPN, IPS, web filtering, and logging	Controls traffic between the internet and internal networks and enforces security policies.
NET-004	Managed Switch	1	48-port Gigabit PoE+ preferred	Connects workstations, server, printers, access points, and other network devices while supporting VLANs.
NET-005	Wireless Access Point	3	Wi-Fi 6/6E class, centrally manageable	Provides business Wi-Fi coverage with separate staff and guest networks.
NET-006	Patch Panel	1	48-port Cat6 patch panel	Provides clean and organized termination for structured network cabling.

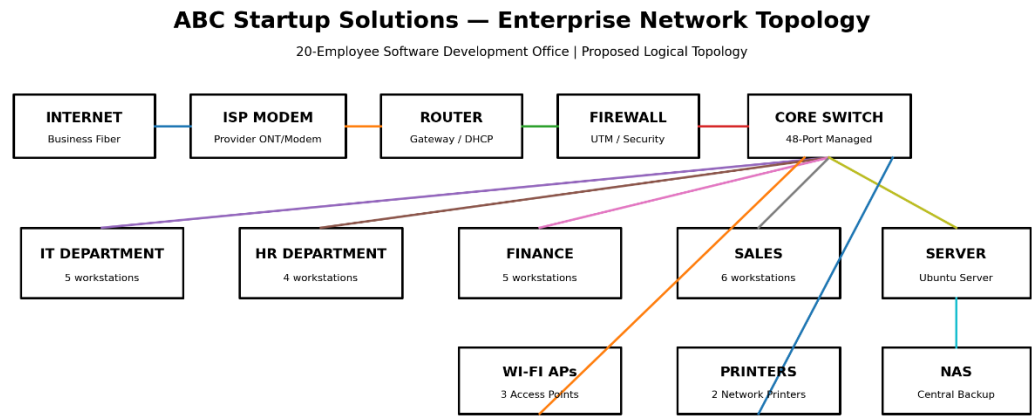
NET-007	CAT6 Ethernet Cable	Approximately 1,000 m	Solid copper, certified Cat6	Used for permanent horizontal cabling and network connections. Final quantity depends on the actual office floor layout.
NET-008	RJ45 Connectors	100	Cat6-compatible	Used for patch cable termination and approved field repairs.

Recommended Network Segmentation

The office should use separate VLANs: VLAN 10 for IT, VLAN 20 for HR, VLAN 30 for Finance, VLAN 40 for Sales, VLAN 50 for servers, and VLAN 60 for guest Wi-Fi. The firewall should control which VLANs can communicate. For example, guest Wi-Fi should have internet access only, while Finance should have restricted access to shared services based on actual business requirements. Management interfaces should be limited to IT administrators.

PART 5 - ENTERPRISE NETWORK DIAGRAM

The proposed topology follows a clear traffic path from the internet provider to the internal network. The ISP modem terminates the service, the router handles gateway functions, and the firewall enforces security policies before traffic reaches the managed switch. The switch then connects the departments, server, printers, and wireless access points.



Logical segmentation recommendation: VLAN 10 IT | VLAN 20 HR | VLAN 30 Finance | VLAN 40 Sales | VLAN 50 Servers | VLAN 60 Guest Wi-Fi

Department-to-department traffic is controlled by firewall rules; guest Wi-Fi is isolated from internal resources.

PART 6 - SYSTEM ADMINISTRATION ROLES

Helpdesk Technician

Responsibilities: first-line technical support, password/account assistance, workstation troubleshooting, software installation, ticket documentation, basic hardware checks, and escalation of complex incidents. Skills: Windows troubleshooting, communication, documentation, basic networking, customer service, and problem isolation. Common tools: ticketing system, remote-support software, Windows Event Viewer, Device Manager, PowerShell, and knowledge bases. Certifications: CompTIA A+, ITIL Foundation, and vendor-specific support credentials are common entry-level options.

Network Administrator

Responsibilities: configure and maintain switches, routers, firewalls, Wi-Fi, IP addressing, VLANs, VPNs, and network monitoring. Skills: TCP/IP, subnetting, routing and switching, wireless networking, security fundamentals, troubleshooting, and documentation. Common tools: Wireshark, ping/tracert, SNMP monitoring platforms, firewall consoles, SSH, and network configuration tools. Certifications: Cisco CCNA, CompTIA Network+, and vendor-specific networking certifications.

Linux System Administrator

Responsibilities: maintain Linux servers, manage users and permissions, install packages, monitor services, manage storage, automate routine tasks, apply updates, and troubleshoot system problems. Skills: Linux command line, Bash scripting, networking, permissions, systemd/services, storage, logs, and security. Common tools: SSH, Bash, systemctl, journalctl, top/htop, rsync, cron, and package managers. Certifications: Linux Foundation certifications, Red Hat certifications, and CompTIA Linux+ are relevant options.

Cloud Administrator

Responsibilities: manage cloud resources, identities, virtual networks, storage, monitoring, backups, access policies, and cost controls. Skills: cloud networking, IAM, virtualization, security, automation, monitoring, and basic scripting. Common tools: AWS/Azure/GCP consoles and CLIs, Terraform, PowerShell, monitoring platforms, and identity-management tools. Certifications: AWS Certified Cloud Practitioner/CloudOps-related credentials, Microsoft Azure Administrator Associate, and Google Cloud Associate Cloud Engineer are examples.

How the Four Roles Work Together

These roles support the same IT environment from different angles. The Helpdesk Technician handles user-facing issues and records incidents, then escalates problems that require deeper access or specialized knowledge. The Network Administrator maintains the connectivity and security of the network. The Linux System Administrator manages server operating systems and the services running on them. The Cloud Administrator manages cloud-hosted resources and identity, storage, and networking services. In a real organization, the roles overlap during incidents, maintenance, deployments, and security events, so clear documentation and escalation procedures are important.

PART 7 - INFRASTRUCTURE RECOMMENDATIONS

Area	Recommendation	Reason
Internet Provider	Use a business-grade fiber connection, preferably around 300–500 Mbps symmetrical at launch, with business support and an SLA. Consider a second ISP as the company grows.	A software company depends on cloud services, Git repositories, video meetings, updates, and remote access. Business support and predictable service are important for daily operations.
Server Specifications	Use a 1U/2U rack server with 8–16 CPU cores, 64 GB ECC RAM, mirrored SSDs for the operating system, redundant storage, dual network interfaces, and remote management.	Provides enough capacity for file services, internal applications, monitoring, and several virtual machines. ECC memory and redundant storage also improve reliability.
Backup Strategy	Follow the 3-2-1 backup approach: three copies of important data, two different storage media, and at least one copy offline or offsite. Use NAS for routine backups and rotate two external drives.	Reduces the impact of accidental deletion, hardware failure, ransomware, or failure of a single backup device.
Security	Implement firewall rules, VLAN segmentation, MFA, least-privilege access, automatic updates, disk encryption, endpoint protection, secure Wi-Fi, logging, and regular security reviews.	A layered security approach reduces unnecessary access and provides several defenses instead of depending on a single security control.
Antivirus / Endpoint Protection	Keep Microsoft Defender enabled on Windows endpoints and configure centralized policies where available. Review alerts and update status regularly.	Provides a consistent endpoint security baseline for Windows computers without immediately requiring a separate antivirus platform.
Password Policy	Require long and unique passwords/passphrases, prohibit password sharing, enable MFA for important accounts, use a password manager for administrators, and disable accounts when employees leave.	Reduces the risk of account compromise and makes user access easier to manage consistently.
Expansion Plan	Reserve approximately 20–30% switch port capacity, maintain spare laptops/workstations, use scalable VLANs, and review	Allows the company to add employees and devices without immediately redesigning the entire network.

	storage and internet capacity every 6–12 months.	
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Implementation Priority

Phase 1: establish internet service, rack/power protection, firewall, router, switch, structured cabling, and Wi-Fi.

Phase 2: deploy employee computers and standard software images.

Phase 3: install and secure the server and NAS, configure backups, and test recovery.

Phase 4: document the environment, review firewall/VLAN rules, and perform a basic security validation before normal operations begin

PART 8 - PERSONAL REFLECTION

Working on this infrastructure planning project helped me understand that system administration starts before a computer is even connected to the network. I used to think of a System Administrator mainly as someone who installs software, fixes computers, or handles technical problems. While those tasks are important, this activity showed me that planning, documentation, security, and resource management are also major parts of the job. The most challenging task was creating the infrastructure from zero. Since the company did not have any existing equipment, I had to decide how many computers, laptops, network devices, access points, storage devices, and backup devices would be reasonable for twenty employees. I also had to think about future expansion instead of designing only for the current number of users. Choosing a 48-port managed switch, additional laptops, multiple access points, and separate backup storage made me realize that infrastructure planning involves looking at both present needs and possible future requirements. Another important lesson was that every hardware and software item should have a purpose. It is not enough to list a router, server, or firewall. A professional infrastructure plan should explain what each device does and why it is appropriate for the organization. The same applies to software. For example, developers need tools for writing and managing code, while the IT team needs administrative and remote-support tools. Business departments need productivity software and reliable access to shared services. Planning is important before deployment because it prevents unnecessary spending and reduces configuration problems. A documented design also gives the IT team something to follow when installing equipment and something to check when troubleshooting. Security is easier to apply when network segments, user access, backups, and administrative responsibilities are considered before the system goes live. Overall, this project helped me see system administration as a combination of technical knowledge, organization, problem solving, and communication. In the future, I can use the same planning approach when designing networks, preparing hardware inventories, documenting systems, and supporting users. It also reminded me that a good infrastructure should not only work on the first day; it should be manageable, secure, documented, and ready to grow with the organization